SDC 2026 · XMUM AI CLUB · TECHNICAL TOOLS DECLARATION

Declared Tools & Resources

Technical Tools Declaration · RREF COLLECTION · Submission deadline 25 October 2026

Instructions followed. Every API, cloud service, AI tool, dataset, model and sponsor resource used to build CollectLoop is declared below. Where a category does not apply we have written "None" or "Not used" rather than leaving the field blank. No API keys or secrets appear in this document.

1 Team and Project Overview

FIELD	RESPONSE
Team Name	RREF COLLECTION
Project / Solution Name	CollectLoop
Name (Leader)	XIE KAIRUI
Email & Phone (Leader)	2173083338@qq.com +86 13906032500
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://rref-collection-sdc.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/Carrey82/rref-collection-sdc

2 Architecture & Technology Stack

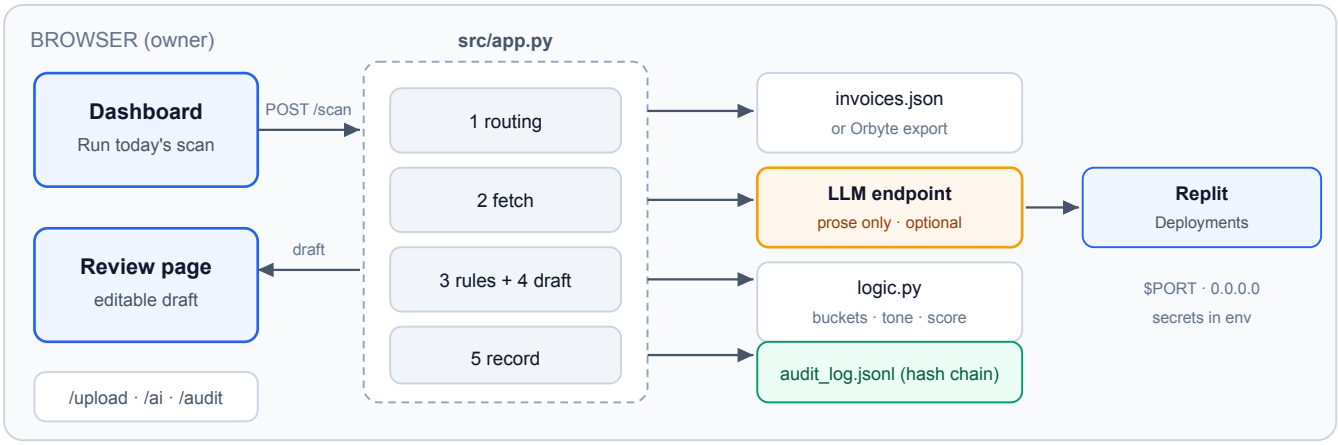
2.1. System Architecture Overview

How the declared components connect end-to-end. [Maximum 200 words]

CollectLoop is one Python process that serves server-rendered HTML; there is no separate frontend build and no client-side JavaScript. Requests enter src/app.py, which routes seven endpoints: dashboard, intake, the AI explainer, review, audit, reset, and the scan trigger.

The operational path has five stages. (1) Trigger: the owner presses "Run today's scan", which posts to /scan. (2) Fetch: store.py loads the invoice book from JSON on disk, or src/orbyte.py normalises an Orbyte export into the same internal shape. (3) Logic: logic.py computes ageing buckets, a priority score and a queue summary, deciding who is chased, who is held and in which tone. (4) Drafting and approval: llm.py builds a fact-restricted prompt and calls an OpenAI-compatible chat endpoint, or falls back to a template; the draft arrives in an editable review page and waits for a human click. (5) Record: store.append_audit writes the decision, the final text and the previous entry's hash to an append-only JSONL log.

Hosting is Replit Deployments; the process binds 0.0.0.0 and reads $PORT. State, audit log and credentials stay outside the repository.



2.2. Languages, Frameworks & Libraries

Main languages, frameworks and third-party libraries the team did not create.

NAME & VERSION	LICENSE	PURPOSE IN PROJECT
Python 3.11	PSF-2.0	Language and runtime for the whole application
Python standard library (`http.server`, `json`, `hashlib`, `urllib`, `html`)	PSF-2.0	Web server, JSON persistence, audit hash chain, model HTTP call, HTML escaping
`unittest` (standard library)	PSF-2.0	45 tests: decision rules, intake validation, security headers
No third-party runtime library	-	Deliberate: the prototype installs nothing, so deployment cannot fail on a dependency
Apache License 2.0	Apache-2.0	Licence we publish our own code under

3 APIs & Cloud Services

3.1. External APIs

Every external API the solution calls. No keys or secrets are included.

API AND PROVIDER	ACCESS TIER	PURPOSE IN PROJECT
OpenAI-compatible Chat Completions API (provider and model configurable; `gpt-4o-mini` class in the default config)	Free tier / pay-as-you-go, configured by the team. No SDC sponsor GenAI credit was issued to or used by this team.	Writes the body of one reminder email from a restricted fact set. Optional: with no key configured the system uses a deterministic template and labels the source in the UI.
Orbyte platform (Domain 2 sponsor) - export schema, not a live API	No credentials issued; Orbyte's Open API is roadmap on their side	We model our internal records on Orbyte's field names and ship an adapter (`src/orbyte.py`) plus a synthetic export sample. **No Orbyte endpoint has been called.** `ORBYTE_MODE=live` deliberately raises NotImplementedError.

3.2. Cloud Platforms & Deployment

Every cloud service used for hosting, storage, databases, authentication or compute.

SERVICE & PROVIDER	PLAN / REGION	PURPOSE IN PROJECT
Replit - Deployments (Autoscale)	Replit Core, sponsor-provided (issued to the team leader)	Hosts the prototype; serves the public `*.replit.app` URL and runs `python3 src/app.py`
Replit Secrets	Replit Core	Holds `LLM_API_KEY` if one is configured; never in the repository
WorkBuddy Sites (static/app hosting)	Free tier	Secondary public URL used if the Replit deployment is unavailable during judging
GitHub	Free (public repository)	Source control, commit history, public judge access, Apache-2.0 licence
RunPod	Not used	No GPU workload: no training, no fine-tuning, no self-hosted inference. `runpod/handler.py` returns `not_implemented` by design.

4 AI Tools, Models & Datasets

4.1. AI Models Used in the Solution

Every model embedded in or called by the application, pre-trained, fine-tuned or self-trained.

MODEL & VERSION	SOURCE / PROVIDER	LICENSE / TERMS	USAGE
OpenAI-compatible chat model (`gpt-4o-mini` class; exact model set by `LLM_MODEL`)	api.openai.com or any compatible provider	Provider terms of service; pre-trained weights, not redistributed	Pre-trained, used as a hosted API. No fine-tuning, no self-hosted weights.
No model is embedded in this build	-	-	In the submitted deployment no API key is configured, so every draft on the live URL comes from the deterministic template. The model call is wired (`src/llm.py`) and the `/ai` page states the current mode openly.

4.2. Datasets

All datasets used for training, fine-tuning, evaluation or in the running application.

DATASET NAME	LICENSE	PERSONAL DATA?	TYPE	SOURCE / URL
`data/invoices.json` - 7 synthetic invoices and customer histories	Self-authored, synthetic	N	Synthetic	Committed in the repository. Every company, person, email and amount is invented, including reserved `.example` domains.
`data/orbyte_export.sample.json` - synthetic Orbyte export	Self-authored, synthetic	N	Synthetic	Models Orbyte's documented field names so the adapter can be reviewed. Contains no real customer data.

4.3. AI-Assisted Development Tools

Did your team use any generative-AI tool to help create code, design, text or other project material? ☑ **Yes** ☐ No

TOOL & VERSION	EXTENT OF USE	USED FOR
WorkBuddy (agentic coding assistant)	Major	Repository scaffolding, HTTP routing and template layer, unit tests, security headers, documentation drafting, and the generation of these submission PDFs
GitHub Copilot (GitHub Student account)	Minor	In-editor completion and small refactors while reviewing the code

How the team reviewed, tested and took responsibility for AI-generated output. [Maximum 100 words]

Every AI-generated change was reviewed against a test suite before acceptance: 45 unittest cases cover the decision rules, intake validation and security headers, and all 45 pass on the submitted commit. Anything a judge could question sits in plain files we can defend line by line - the ageing buckets and tone ladder in src/logic.py, the

prompt in src/llm.py. We also rejected two AI suggestions: a Flask-based framework layer that broke deployment, and letting the model choose urgency, because it gave different answers for identical inputs. The team owns every claim here.

5 Sponsor-Provided Resources

Did your project use any official SDC sponsor-provided resource? ☑ **Yes** ☐ No

RESOURCE	SPONSOR	HOW IT WAS USED
Replit Core (cloud platform and deployment credits)	Orbyte / SDC 2026 sponsor agreement (issued per participant to the team leader)	Hosts the deployed prototype and the public live URL; secrets storage; the deployment that judges open
SDC Phase 1 workshops (GenAI API, database architecture, cloud deployment)	SDC / XMUM AI Club	Shaped the model layer, the audit design and the deployment approach; no workshop code is embedded
Orbyte platform (Domain 2 sponsor)	SDC Domain 2 sponsor	Domain brief, data model and field naming. No API credentials were issued and no Orbyte endpoint was called.

Was the sponsor resource essential to core functionality, or could it be replaced? [Maximum 100 words]

The Replit deployment was essential at submission time: it is the address judges open, and it satisfies Handbook 10.6 proof of use. It is not essential to the product. There are no proprietary calls - the application is one Python process with zero dependencies, and porting it to any container host took minutes, which is exactly what we did when the first Replit publish failed: the same commit now also runs on a second public host. Orbyte matters as a data contract, not as a runtime dependency, so its absence constrains the integration, not the prototype.

6 Security, Privacy & Responsible Use

6.1. Data Handling & Privacy

What user or personal data is collected, stored or sent to third parties, how is it protected, and how long is it kept? [Maximum 150 words]

No real user data is collected, stored or transmitted. The invoice book, customer names, emails and amounts in data/invoices.json are synthetic; emails use the reserved .example domain. The intake form at /upload accepts six typed fields and stores them in a local JSON file on the hosting instance. There is no account system, no login, no analytics and no application cookies. When a model is configured, only the fact set needed to draft one email is sent to the provider - invoice number, amount in MYR, due date, days overdue and the customer's recorded promise, which is already synthetic; no free-text personal notes, and no API key in the browser. The append-only audit log stores the decision, the final draft text and a hash chain on the same instance. Retention: until the demo is reset (/reset or a redeploy), after which the runtime state is gone.

6.2. Known Limitations & Risks

Main technical limitations of the AI components and any mitigation implemented. [Maximum 150 words]

Four limitations are known and stated rather than hidden. Hallucination: the model only writes prose from a closed fact set and cannot invent amounts, dates or legal threats; a human approves every send, and sending itself is deliberately unwired. Provider dependency and rate limits: the call has a timeout and falls back to a template, and the interface labels which source produced the draft, so the demo degrades visibly instead of white-screening. Explainability: tone and priority are deterministic rules in one file, not model output, so any decision can be reproduced and argued. Data layer: JSON files are not concurrency-safe and are single-tenant, which is acceptable

for one SME's book and not for multi-user deployment; the Orbyte integration is an adapter with no live endpoint, and no OCR or QR intake is implemented. Synthetic data means the measured business impact is illustrative, not evidence.

7 Compliance & Readiness

Tick all boxes that apply before submitting.

☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☑ Any use of official SDC sponsor resources has been declared in Section 5.

☑ The prototype described here is accessible to judges through the links provided in Section 1.

☑ All fields in this template have been completed within the word limits stated.

8 Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model and sponsor resource used in the project has been declared.

Signature - XIE KAIRUI, Team Leader, RREF COLLECTION

Date - 24 October 2026